5.0 Jointed Rods

In previous chapters 4.1 and 4.2 we considered the action of coplanar forces acting on a single rigid body In this chapter we shall consider the action of coplanar forces acting on two or more rigid bodies, specially on number of heavy rods jointed together to form a frame.

We will consider the equilibirium of rods under the action of their weight, any external forces applied and forces exerted on their ends by hings (joints)

5.1 Types of simple joints

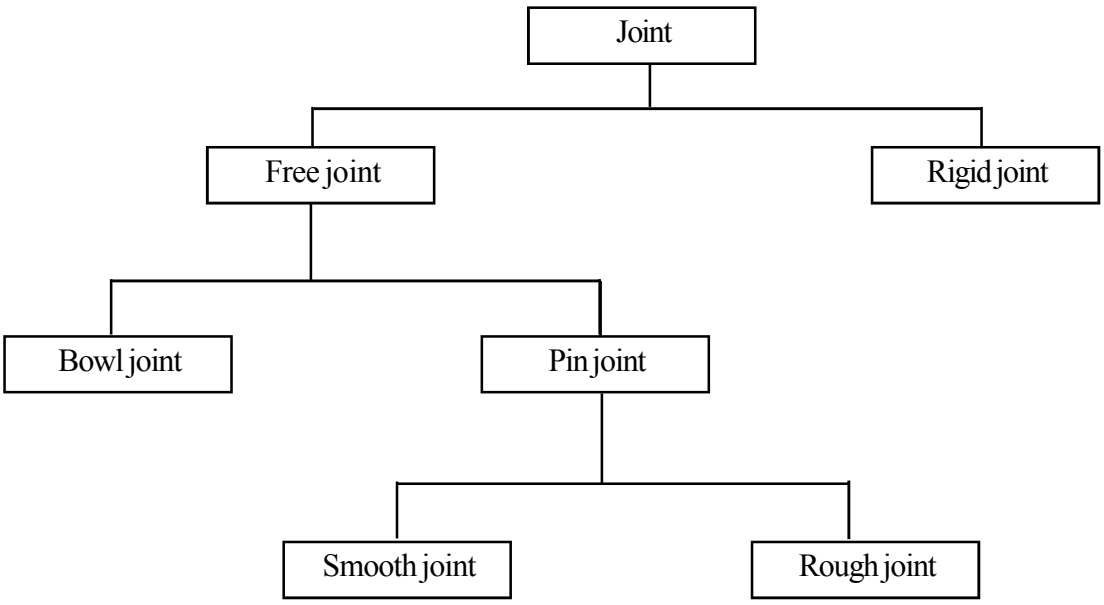
(i) Rigid Joint

When two rods are jointed together such a way that they cannot be seperated or turn about one another at the joint, the joint is rigid joint.

(ii) Pin Joint

When two rods are jointed by a light pin such a way that they can turn at the joint, the joint is pin joint if they can turn freely at the joint, the joint is a smooth pin joint and, if free turn is not possible the joint is a rough joint.

We shall consider the frames with smooth pin joints in this chapter.

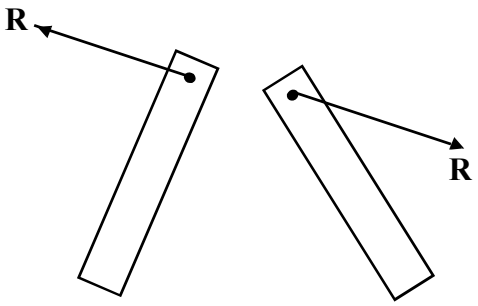


5.2 Rigid joint

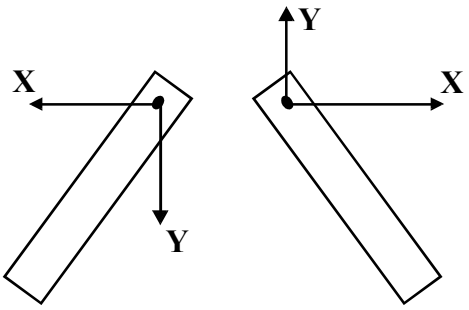
If the shape of a body obtained by joining two or more bodies together cannot be changed by external forces then the joint is said to be a rigid joint.

Force at a smooth joint (Pin joint)

The joints are shown seperately to show the reaction at the joint. The reaction at the joints will be equal and opposite. To find R easily the components of R are shown as follows.



X, Y are the horizontal and vertical components of R. R is the resultant of X and Y, and passes through the pin joint.

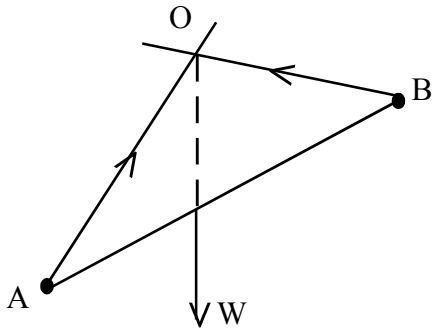


The light pin is assumed to be a small smooth pin of circular rim, joins the rod by passing through the rods. As the pin is smooth the reaction on contact is perpendicular to it and for rods. Since the pin is in equilibirium under these two forces, they are equal opposite in direction and have the sameline of action. Therefore the reaction on each rod is equal and opposite and have the same line of action an each joints.

For conveneience we resolve the reaction into two perpendicular components when we need of it.

Note :

When a heavy rod is joined at its ends to another rod, the reaction by joints on the rod cannot lie along the rod, since the rod is acted by three forces.



For equilibrium forces should meet at one point O which cannot lie on AB.

If the rod is light, it is acted by the two reactions only, so that they always lie along the rod to balance each other.

When a framework is symmetric about an axis identical set of forces will act on both sides.

Instructions to solve problems

(i) Correct diagram has to be drawn with geometrical data.

(ii) Forces should be marked correctly.

(iii) Necessary equations should be obtained to find the unknown forces.

(iv) To find the reactions at a joint the force at the joint should be divided into two components and to be marked.

(If there is axis of symmetry, it should be stated and the results can be used)

Note :

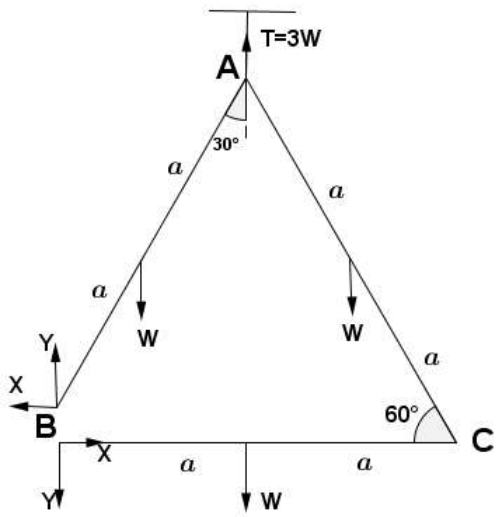
A framework must be rigid. To make a framework of n jonts (n >3) to be rigid it is necessary to have (2n-3) rods.

A framework with more than (2n-3) rods will make the framework over rigid.

5.3 Worked examples

Example 1

Three uniform equal rods of length $2a$ and weight W are freely jointed at their end points and the frame ABC is suspended from the joint A. Find the magnitude and direction of the reaction at B on AB.



Consider the equilibrium of BC

Taking moments about C for BC

Cm $\quad W.a + Y.2a = 0$

$$2Y + W = 0 \quad ; Y = -\frac{W}{2}$$

Consider the equilibrium of AB.

Taking momements about A for AB

Am $\quad Y(2a \sin 30°) + X(2a \cos 30°) - W(a \sin 30°) = 0$

$$2Y + 2X \cot 30° = W$$

$$2Y + 2\sqrt{3}X = W$$

$$-W + 2\sqrt{3}X = W \quad ; X = \frac{W}{\sqrt{3}}$$

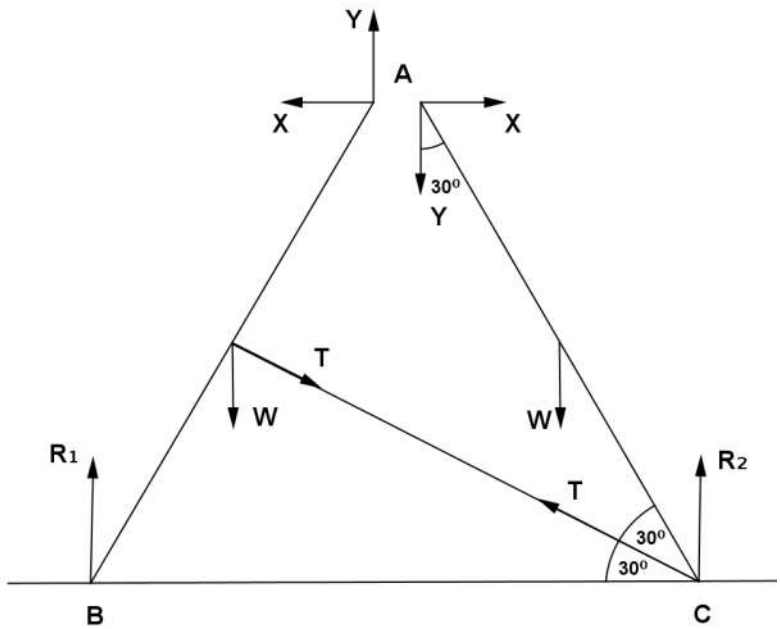
$$R = \sqrt{X^2 + Y^2} = \sqrt{\frac{W^2}{3} + \frac{W^2}{4}} = \sqrt{\frac{7}{12}}W$$

$$\tan\theta = \frac{Y}{X} = \frac{\sqrt{3}}{2} ; \quad \therefore \theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

Magnitude of the reaction at B $= \sqrt{\frac{7}{12}}$ W ; R makes an angle θ with CB where $\theta = \tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Example 2

Two uniform rods AB, AC each of length $2a$ and weight W are smoothly jointed at A. The rods are in equilibrium in a vertical plane with B and C lying on a smooth horizontal plane and C is connected to the midpoint of AB by an inextensible string and $B\hat{A}C = 60°$. Find the tension in the string and the reaction at A.



$AB = AC$; $B\hat{A}C = 60°$

Therefore ABC is an equilateral triangle.

For equilibrium of AB and AC,

Resolving vertically

$\uparrow \quad R_1 + R_2 - 2W = 0$; $R_1 + R_2 = 2W$ ..... ①

Taking moment about C

Cm $\quad -R_1 . 4a \cos 60° + W . a \cos 60° + W . 3a \cos 60° = 0$ ..... ②

$R_1 = W$ and $R_2 = W$

For equilibrium of AC,

Am $\quad -W.a \cos 60° - T . a + R_2 . 2a \cos 60° = 0$ ..... ③

$$-\frac{W}{2} - T + W = 0; \quad T = \frac{W}{2}$$

For equilibrium of AC, moment about A

Resolving horizontally,

$$\rightarrow X - T \cos 30° = 0 \ ; \ X = T \cos 30° = \frac{\sqrt{3}W}{4}$$

Resolving vertically,

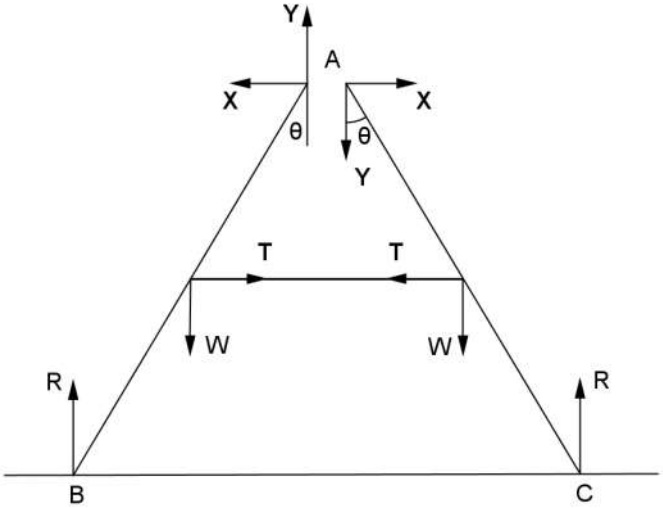
$$\uparrow R_2 - Y - W + T \sin 30° = 0$$

$$Y = R_2 - W + \frac{T}{2} = \frac{W}{4}$$

Hence reaction at A is $\sqrt{X^2 + Y^2} = \sqrt{\frac{3W^2}{16} + \frac{W^2}{16}} = \frac{W}{2}$

Example 3

Two uniform equal rods AB, AC each of weight W are smoothly jointed at A. The ends B and C rest on a horizontal smooth plane and the frame ABC is kept in a vertical plane. The equilibrium is maintained by connecting midpoints of AB and AC by an inextensible string. If $B\hat{A}C = 2\theta$, find the tension in the string and the magnitude of the reaction at A on AB.



Let AB = AC = $2a$

For the equilibrium of AB and AC,

Resolving vertically,

$\uparrow 2R - 2W = 0$

$R = W$ .....①

For equilibrium of AB,

Resolving vertically,

$\uparrow R + Y - W = 0$

$W + Y - W = 0$; $Y = 0$ .....②

Resolving horizontally,

$\rightarrow T - X = 0$; $T = X$ ..... ③

Taking moment about A for equilibrium of AB,

Am $\quad T.a\cos\theta + W.a\sin\theta - R.2a\sin\theta = 0$

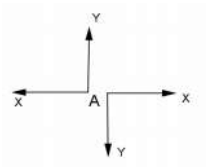
$$T = \frac{(2W - W)\sin\theta}{\cos\theta} = W\tan\theta \text{ .....④}$$

Reaction at A is $W\tan\theta$

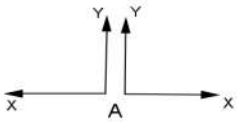
Note :

In the above example the system is symmetrical about the vertical axis through A

Now the reaction at A is given by



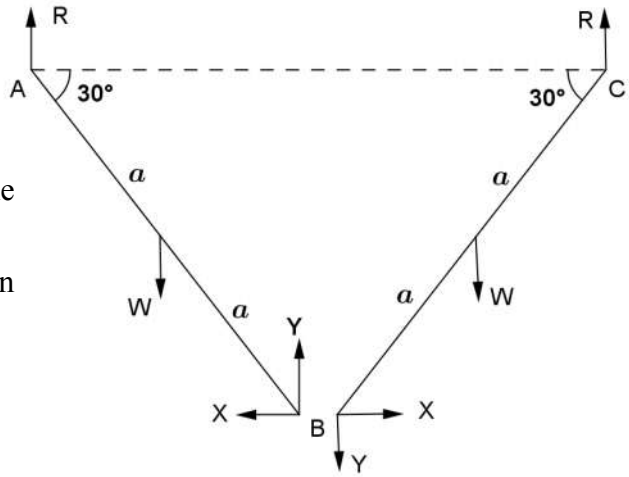
Since the system is symmetri about the vertical axis through A, the forces should be as given below.



Hence Y = 0

Example 4

AB, BC are two uniform rods each of lengh $2a$ and weight W, smoothly hinged at B, and the frame ABC is suspended from the points A and C at the same horizontal level. The systerm is in a vertical plane and each rod makes 30° with the horizontal. Find the reaction at the joint B.



The system is symmetrical about the vertical line through B.

Therefore the vertical component (Y) of the reaction at B is zero (Y=0)

For the equilibrium of AB

By taking moments about A

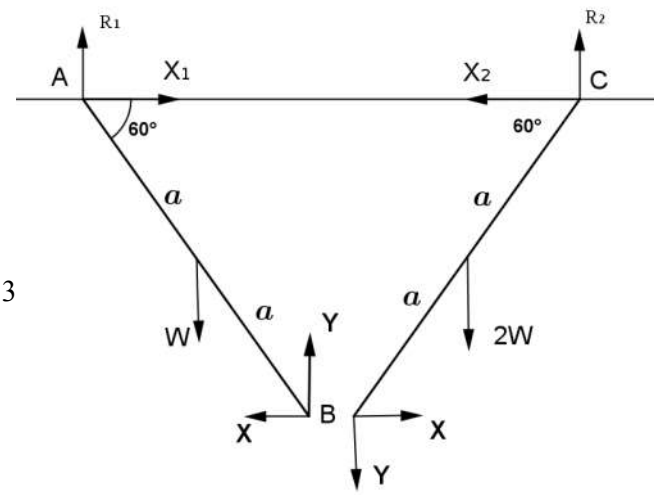
$$\text{Am} \quad -X\,.\,2a\sin 30° + Y\,.\,2a\cos 30° - Wa\cos 30° = 0$$

$$-X\,.\,2a\sin 30° = W.a\cos 30°$$

$$X = -\frac{\sqrt{3}W}{2}$$

Example 5

AB, BC are two equal uniform rods each of length $2a$ and weight W and $2W$ respectively. The rods are smoothly jointed at B and the frame ABC is suspended from A and C at the same horizontal level. The system is in the vertical plane and each rod makes 60° with the horizontal. Find the magnitude and the direction of the reaction at the joint B on AB.



For equilibrium of the system

Resolving horizontally,

$$\rightarrow X_1 - X_2 = 0\ ;\ \ X_1 = X_2$$

Resolving vertically

$$\uparrow R_1 + R_2 - 3W = 0 \quad ;\ \ R_1 + R_2 = 3$$

For AB and AC moment about A

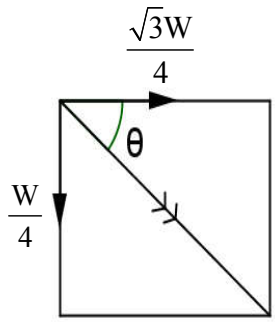
$$\text{Am} \qquad R_2.2a - W.\frac{a}{2} - 2W.\frac{3a}{2} = 0$$

$$2R_2 = \frac{7W}{2}\ ;\ R_2 = \frac{7W}{4} \text{ and } R_1 = \frac{5W}{4}$$

For equilibrium of BC,

Resolving vertically $\uparrow$ $R_2 - 2W - Y = 0\ ;\ Y = R_2 - 2W = \frac{7W}{4} - 2W = -\frac{W}{4}$

For equilibrium of BC



Cm $\quad X \,.\, 2a \sin 60° + Y.\, 2a \cos 60° + 2W.\, a \cos 60° = 0$

$$X.\, 2a \sin 60° - \frac{W}{4}.\, 2a \cos 60° + 2W.\, a \cos 60° = 0$$

$$X = -\frac{\sqrt{3}W}{4}$$

$$R = \sqrt{\frac{3W^2}{16} + \frac{W^2}{16}}$$

$$R = \frac{W}{2}$$

$$\tan\theta = \frac{\frac{W}{4}}{\frac{\sqrt{3}W}{4}} = \frac{1}{\sqrt{3}}$$

$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= \frac{\pi}{6}$$

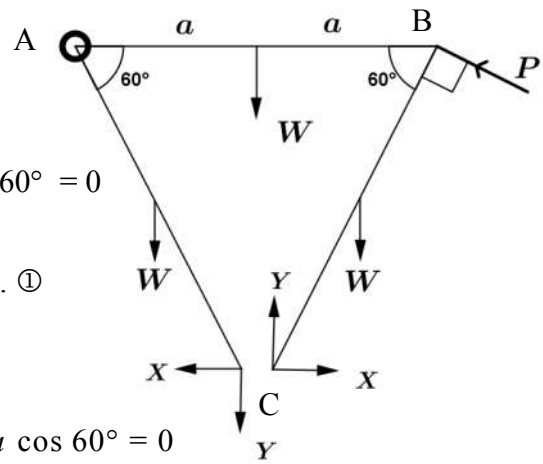
Example 6

Three uniform equal rods AB, BC, AC each of length $2a$ and weight W are smoothly jointed at their ends to form an equilateral triangle. The frame is freely hinged at A in a vertical plane. The triangle is kept in equilibrium with AB as horizontal and C is below AB by a force at B perpendicular to BC by a force P at B perpendicular to BC. AB is horizontal and C is below AB. Find the value of P. Also find the horizontal and vertical components of the horizontal and vertical components of the reaction at C.

By taking moments about A for the system

Am $\quad -W.a \cos 60° - W.a - W\,(2a - a \cos 60°) + P.\, 2a \cos 60° = 0$

$$P = 3W$$



By taking moment about A for equilibrium of AC,

Am $\quad X.\, 2a \sin 60° + Y.\, 2a \cos 60° + W.\, a \cos 60° = 0$

$$\Rightarrow X + \frac{Y}{\sqrt{3}} = -\frac{W}{2\sqrt{3}} \quad \text{.....} \quad ①$$

By taking moments about B for equilibrium of BC,

Bm $\quad X \,.\, 2a \sin 60° - Y \,.\, 2a \cos 60° + W \,.\, a \cos 60° = 0$

$$\Rightarrow X - \frac{Y}{\sqrt{3}} = -\frac{W}{2\sqrt{3}} \quad \text{.....} \quad ②$$

From ① and ②

$$Y = 0$$

$$X = -\frac{W}{2\sqrt{3}}$$

Reaction at C is $\frac{W}{2\sqrt{3}}$

Example 7

Two uniform rods AB and BC each of length $2a$ and weights 2W, W respectively are smoothy hinged at B. The mid points of the rods are connected by a light in inelastic string. The system in a vertical plane with other ends A and C lie on a smooth horizontal table. If $A\hat{B}C = 2\theta$ show that the tension in the string is $\frac{3W}{2}\tan\theta$. Find the magnitude and direction of the reaction at B.

For the equilibrium of the system,

By taking moments about C

$$Cm \quad W.\,a\sin\theta + 2W.\,3a\sin\theta - R.\,4a\sin\theta = 0$$

$$R = \frac{7W}{4}$$

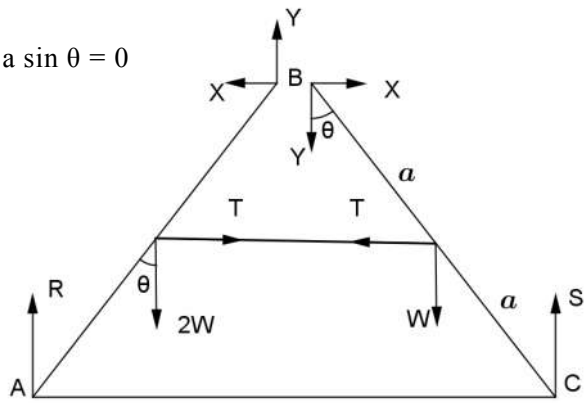
For equilibrium of AB , taking moment about B

$$Bm \quad T.\,a\cos\theta + 2W.\,a\sin\theta - R.\,2a\sin\theta = 0$$

$$T = -2W\tan\theta + 2R.\tan\theta$$

$$T = -2W\tan\theta + \frac{7W}{2}\tan\theta$$

$$T = \frac{3W}{2}\tan\theta$$



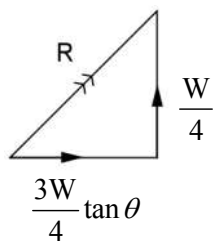
For equilibrium of AB

Resolving horizontally,

$$\rightarrow\ T - X = 0\ ;\ X = T = \frac{3W}{2}\tan\theta$$

Resolving vertically,

$$\uparrow\ Y + R - 2W = 0$$



$$Y = 2W - \frac{7W}{4} = \frac{W}{4}$$

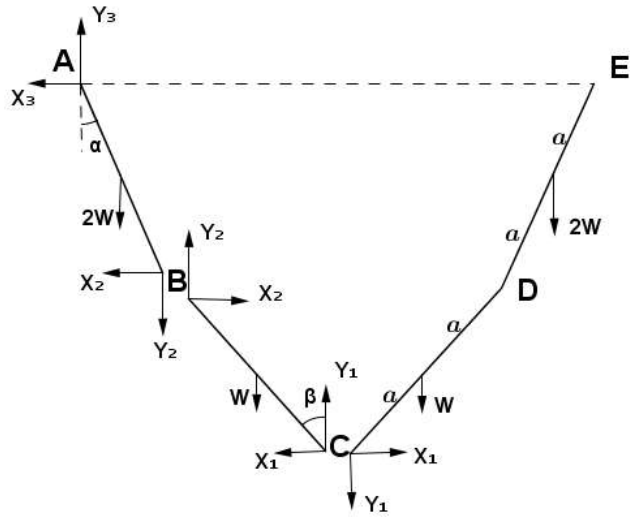
$$R = \sqrt{X^2 + Y^2}$$

$$= \sqrt{\frac{9W^2}{4}\tan^2\theta + \frac{W^2}{16}}$$

$$= \frac{W}{4}\sqrt{1 + 36\tan^2\theta}$$

Example 8

AB, BC, CD, DE are four uniform equal rods of length $2a$, smoothly jointed at B, C and D. The weights of AB, DE are $2W$ each and the weights of BC, CD are W each. The system is suspended from A and E at the same horizontal level. AB and BC make α, β with the vertical respectively. Show that the reaction at C is horizontal and the magnitude is $\frac{W}{2}\tan\beta$. Show also that $\tan\beta = 4\ \tan\alpha$.



The system is symmetrical about the vertical axis through C

Therefore the vertical component of the reaction at C is zero. $Y_1 = 0$

For the equilibrium of BC

Resolving the forces horizontally

$$\rightarrow X_1 - X_2 = 0$$

$$X_1 = X_2$$

Resolving the forces vertically

$$\uparrow Y_1 + Y_2 - W = 0$$

$$Y_2 = W$$

moment about B

Bm $\quad -X_1 \,.\, 2a\cos\beta - W \,.\, a\sin\beta = 0$

$$X_1 = -\frac{W}{2}\tan\beta$$

For equilibrium of AB,

Am $\quad -X_2 \,.\, 2a\cos\alpha + 2W \,.\, a\sin\alpha + Y_2 .\ \ 2a\sin\alpha = 0$

$$X_2 = -2W\tan\alpha$$

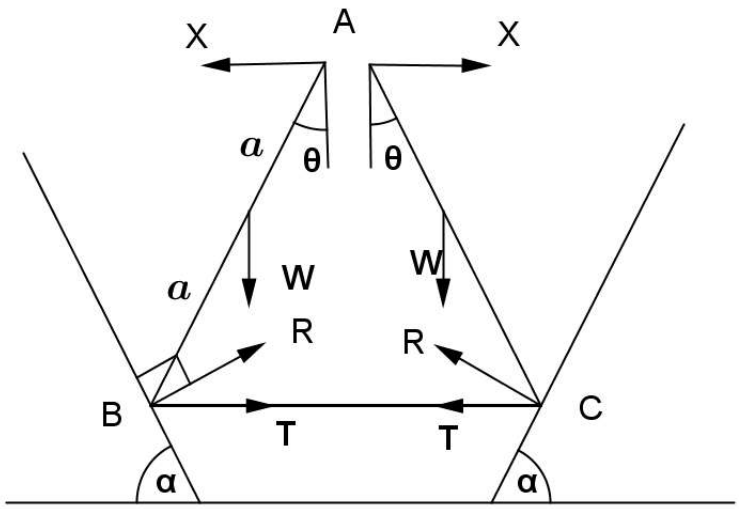
$$X_1 = X_2$$

$$\frac{W}{2}\tan\beta = 2W\tan\alpha$$

$$\tan\beta = 4\tan\alpha$$

Example 9

Two equal uniform rods AB and AC each of weight W are freely jointed at A, and the ends B and C are connected by a light inextensible string. The rods are kept in equilibrium in a vertical plane with the ends B and C on two smooth planes each of which inclined at an angle α to the horizontal; BC being horizontal and A being above BC. Find the reaction at B. If $\tan\theta > \tan 2\alpha$, where $B\hat{A}C = 2\theta$ then show that the tension in the string is $\frac{1}{2}W(\tan\theta - 2\tan\alpha)$. Find also the reaction at the joint A.



Let $2a$ be the length of each rod.

The system is symmetrical about the vertical axis through A.

Hence the vertical component of the reaction at A is zero.

For equilibrium of the system

Resolving vertically,

$$\uparrow\ 2R\cos\alpha - 2W = 0\ ;\ R = W\cos\alpha$$

For equilibrium of AB, Taking moment about A

$$Am \quad T.\,2a\cos\theta + R\sin\alpha\,.\,2a\cos\theta + W\,.\,a\sin\theta - R\cos\alpha\,.\,2a\sin\theta = 0$$

$$T = \frac{W}{2}(\tan\theta - 2\tan\alpha)$$

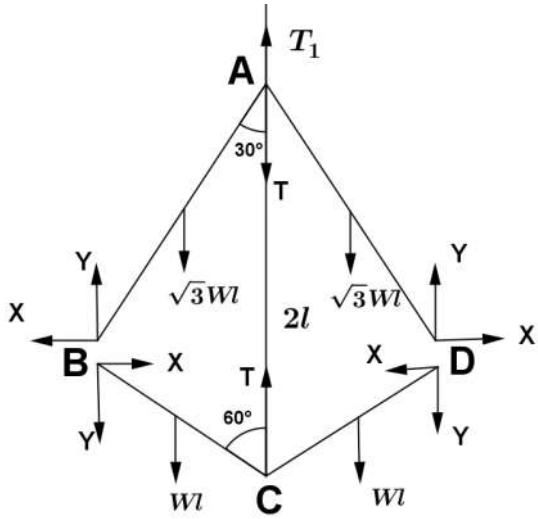
For equilibrium of AB,

$$Bm \quad X\,.\,2a\cos\theta - W\,.\,a\sin\theta = 0$$

$$X = \frac{W}{2}\tan\theta$$

Example 10

AB, BC, CD and AD are four uniform rods having lengths AB = AD = $\sqrt{3}\ell$ and BC = DC = ℓ and are smoothly jointed at their ends to form a frame ABCD. The rods have weights W per unit length. The joints A and C are connected by an inelastic string of length 2ℓ. The frame is suspended in a vertical plane from A. Show that the tension in the string is $\frac{W\ell}{4}\left(\sqrt{3}+5\right)$



Method 1

$AB^2 + BC^2 = 3\ell^2 + \ell^2 = 4\ell^2 = AC^2$

Therefore, $A\hat{B}C = 90°$, $B\hat{A}C = 30°$, $B\hat{C}A = 60°$

The system is symmetrical about AC. Hence reactions at B and D are same.

For equilibrium of AB, Taking moment about A

$$\text{Am} \quad X.\sqrt{3}\ell\cos 30° + Y.\sqrt{3}\ell\sin 30° - \sqrt{3}\ell W.\frac{\sqrt{3}}{2}\ell\sin 30° = 0$$

$$X.\cot 30° + Y = \frac{\sqrt{3}}{2}\ell W$$

$$\sqrt{3}X + Y = \frac{\sqrt{3}}{2}\ell W \quad \text{.....} \quad ①$$

For equilibrium of BC , moment about C

$$\text{Cm} \quad Y.\ell\sin 60° + W\ell.\frac{\ell}{2}\sin 60° - X.\ell\cos 60° = 0$$

$$Y + \frac{W\ell}{2} = \frac{X}{\sqrt{3}}$$

$$X = \sqrt{3}Y + \frac{\sqrt{3}W\ell}{2} \quad \text{.....} \quad ②$$

Substitute ① and ②

$$Y + \sqrt{3}X = \frac{\sqrt{3}W\ell}{2}$$

$$Y + \sqrt{3}\left(\sqrt{3}Y + \frac{\sqrt{3}W\ell}{2}\right) = \frac{\sqrt{3}W\ell}{2}$$

$$4Y + \frac{3W\ell}{2} = \frac{\sqrt{3}W\ell}{2}$$

$$Y = \frac{W\ell}{8}\left(\sqrt{3}-3\right)$$

for equilibrium of BC and CD

$$\uparrow\ T - 2Y - 2W\ell = 0$$

$$T = 2Y + 2W\ell$$

$$T = 2\frac{W\ell}{8}\left(\sqrt{3}-3\right) + 2W\ell$$

$$T = \frac{W\ell}{4}\left(\sqrt{3}+5\right)$$

or For BC and CD take moments about D

Method 2

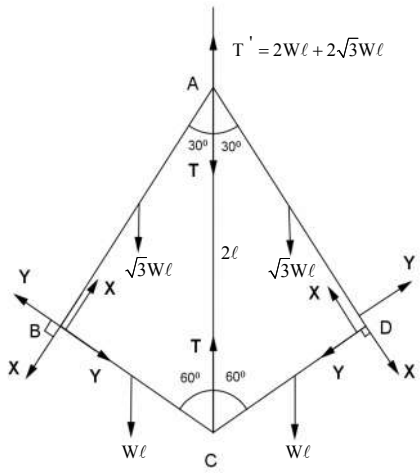
$AB^2 + BC^2 = 3\ell^2 + \ell^2 = 4\ell^2 = AC^2$

$A\hat{B}C = 90^0$, $B\hat{A}C = 30^0$, $B\hat{C}A = 60^0$

By symmetry reactions at B and D are same.

The components of the reaction at B are taken along BA and BC, since $A\hat{B}C = 90^0$

For the equilibrium of the rod AB,



Am $\quad \sqrt{3}W\ell.\ \frac{\sqrt{3}\ell}{2}\sin 30° - Y.\sqrt{3}\ell = 0$

$$Y = \frac{\sqrt{3}W\ell}{4}$$

For the equilibrium of BC,

Cm $\quad W\ell.\ \frac{\ell}{2}\sin 60 - X.\ell = 0$

$$X = \frac{\sqrt{3}W\ell}{4}$$

For the equilibrium of BC and CD, Resolving vertically

$$T - 2W\ell + 2X\cos 30° - 2Y\cos 60° = 0$$

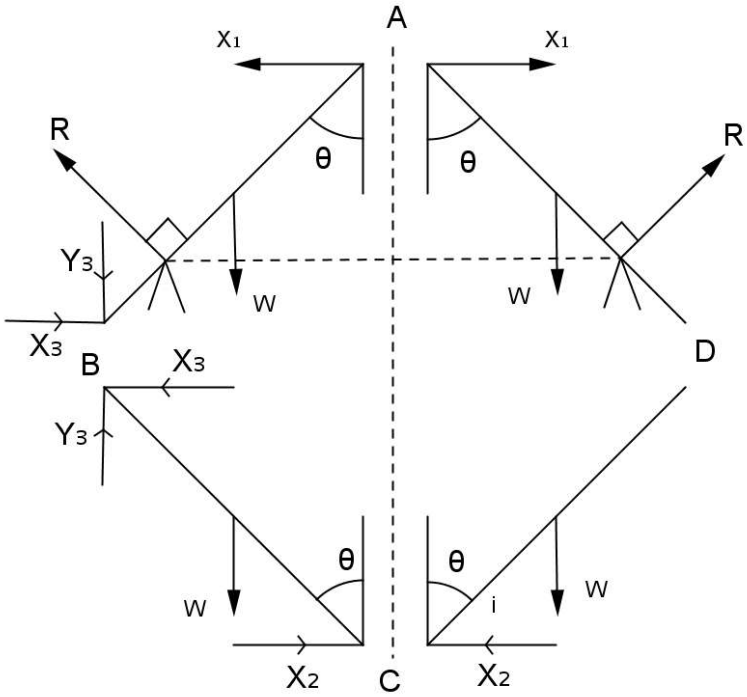
$$T = 2W\ell + 2Y\cos 60° - 2X\cos 30°$$

$$= 2W\ell + \frac{\sqrt{3}W\ell}{4} - \sqrt{3}.\frac{\sqrt{3}W\ell}{4}$$

$$= \frac{W\ell}{4}\left(\sqrt{3}+5\right)$$

Example 11

Four uniform rods AB, BC, CD, DA each of length $2a$ and weight W are freely hinged at their ends, and rest with the upper rods AB, AD in contact with two smooth pegs in the same horizontal line at a distance $2c$ apart. If the inclination of the rods to the vertical is θ, determaine the horizontal and vertical components of the reaction at B and prove that $c = 2a\sin^3\theta$.



The system is symmetrical about AC. Therefore, the vertical components of the reaction at A and C are zero.

For equilibrium of the system,

Resolving vertically,

$$\uparrow 2R\sin\theta - 4W = 0$$

$$R = \frac{2W}{\sin\theta} \text{ ..... ①}$$

For equilibrium of BC,

Bm $\quad X_2 . 2a\cos\theta - W.a\sin\theta = 0$

$$X_2 = \frac{W\tan\theta}{2} \text{ ..... ②}$$

Resolving horizontally,

$$\rightarrow X_2 - X_3 = 0;\ X_3 = X_2 = \frac{W\tan\theta}{2} \text{ .....③}$$

Resolving vertically

$$\uparrow Y_3 - W = 0\ ;\ Y_3 = W \text{ .....④}$$

For equilibrium of AB,

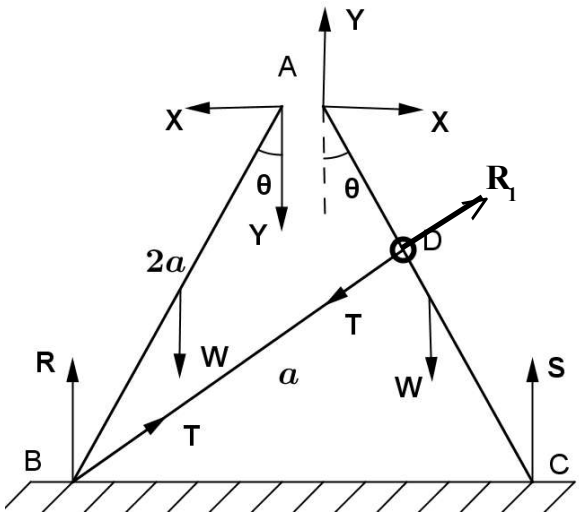
Am $\quad -R.\frac{c}{\sin\theta} + Wa.\sin\theta + Y_3.2a\sin\theta + X_3.2a\cos\theta = 0$

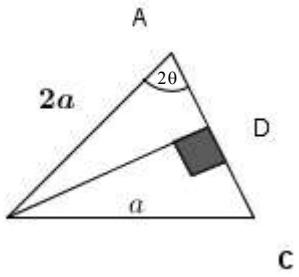
$$-\frac{2W.c}{\sin^2\theta} + W.a\sin\theta + W.2a\sin\theta + \frac{W}{2}.2a\sin\theta = 0\ ; \quad c = 2a\sin^3\theta$$

Example 12

Two equal uniform rods AB, AC each of length $2a$ and weight W are smoothly jointed at A. BD is a weightless bar of length a, smoothly jointed at B and fastened at D to a small smooth light ring sliding on AC. The system is in equilibrium in a vertical plane with ends B and C resting on a horizontal plane. Show that the magnitude of the reaction at A is equal to $\frac{W}{12}\left(3\sqrt{2}-\sqrt{6}\right)$. Also show that the magnitude of the reaction at A is equal to the stress on BD and it makes an angle 15° with the horizontal.

Find the point where the line of action meets BC.





$A\hat{D}B = 90°$

$\sin 2\theta = \frac{1}{2}$

$2\theta = \frac{\pi}{6}$

$\theta = \frac{\pi}{12}$

For the equilibrium of the ring $R_1 = T$ and R_1 is perpendicular to AC, so T is perpenticular to AC

For the system

Resolve the forces vertically

$\uparrow \quad R + S = 2W$

By taking moment about C

$\circlearrowleft m \quad W.a \cos 75° + W.\,3a \cos 75° = R.\,4a \cos 75° = 0$

$\Rightarrow R = W$

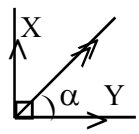
$R = S = W$

Consider the equilibrium of the rod AC

By taking moments about A for AC

$\circlearrowleft m \quad T.\,a\sqrt{3} + W.a \sin 15° - W.2a \sin 15° = 0$

$$T = \frac{W \sin 15°}{\sqrt{3}} = \frac{W}{12}\left(3\sqrt{2}-\sqrt{6}\right)$$



For the rod AB resolve the forces horizontally and vertically

$\rightarrow \quad X = T \cos 15° \;; \qquad \uparrow \quad Y = T \sin 15° \;;$

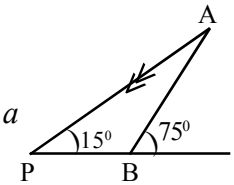
$A = \sqrt{X^2 + Y^2} = T \;; \qquad \tan\alpha = \frac{X}{Y}$

$= \tan 15°$

$\alpha = 15^0$

Using sine rule in ΔABP $\quad \frac{BP}{\sin 60°} = \frac{AB}{\sin 45°} \Rightarrow BP = \frac{2a\,.\sin 60°}{\sin 15°}$

$$BP = \frac{2a.\frac{\sqrt{3}}{2}}{\frac{\sqrt{3}-1}{2\sqrt{2}}} = \frac{2\sqrt{6}}{\sqrt{3}-1} \quad \Rightarrow BP = \left(3\sqrt{2}+\sqrt{6}\right)a$$



5.4 Exercises

1. Two uniform rods AB and AC of equal length are freely hinged at B. The weights of AB and BC are W_1 and W_2 respectively. The system freely hangs from fixed points B and C at the same level and BC = $2a$. If the depth of A below BC is a, find the horizontal and vertical components of the reaction at A.

2. Two equal uniform rods AB, BC each of weight W are smoothly jointed at B and their midpoints are connected by an inextensible string. The string has length such as when it is taut $A\hat{B}C$ makes 90°. The system is suspended from A freely while the string is taut. Show that the inclination of AB with the vertical at equilibrium is $\tan^{-1}\left(\frac{1}{3}\right)$ and the tension in the string is $\frac{3W}{\sqrt{5}}$. Also find the reaction on BC and show that it is in the direction of BC.

3. Two uniform equal rods AB, AC of length $2a$ and weight W, smoothly jointed at A lie symmetrically on the curved surface of a right circular cylinder whose axis is fixed horizontally. If each rod makes an angle θ with the horizontal and r is the radius of the cylinder, show that $r = a \operatorname{cosec} \theta \cos^3 \theta$. Find also the reaction at A.

4. AB, BC and AC are three uniform equal rods smoothly jointed at ends A , B and C. AB and AC are each of weight W and the weight of BC is $2W$. The frame hangs freely from C. Show that BC makes an angle $\tan^{-1}\left(\frac{4}{\sqrt{3}}\right)$ with the horizontal. Find also the reaction at A and B.

5. Two uniform equal rods AOB and COD each of weight W are freely jointed at O, AO = CO = a, and BO = OD = $3a$. At equilibrium B and D rest on a horizontal plane and B, D are connected by an inextensible string of length $3a$. The system lies in equilibrium in a vertical plane. Show that the tension in the string is $\frac{2\sqrt{3}\,W}{9}$ and find the reaction at O.

6. Two uniform equal rods AB and AC of weight $2W$ and W, respectively, are smoothly jointed at A. B and C are fixed to a horizontal log. Find the horizontal and vertical components of the reaction at A. If the reaction at B and C are perpendicular to each other and $A\hat{B}C = \alpha$, show that $3\cot\alpha = \sqrt{35}$.

7. Three uniform equal rods OA, AB and BC each of length $2a$ and weight W are freely jointed at A and B. The end O is hinged to a fixed point and a horizontal force P is applied to BC at C and BC makes an angle 45° to the horizontal. Find P in terms of W. Show that the reaction at O is $\frac{\sqrt{37}\,W}{2}$. Show also that C is at a horizontal distance $2a\left[\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{10}}+\frac{1}{\sqrt{26}}\right]$ from the vertical through O.

8. Two equal uniform rods AB, BC each of length a and weight W are smoothly jointed at B. The rod AB is free to rotate about the point at which A is hinged. A small light ring is attached to C which is free to slide along another fixed rod through A. The fixed rod is inclined downwards, making an angle α to the horizontal. If the system is in equilibrium show that

   (i) $\tan B\hat{A}C = \frac{1}{2}\cot\alpha$

   (ii) The horizontal component of the reaction at B is $\frac{3W}{8}\sin 2\alpha$.

9. Four uniform equal rods AB, BC, CD and AD each of weight W are smoothly jointed at their ends to form a rhombus ABCD and hangs from A. The system is maintained in the shape of a square connecting the midpoints of BC and CD by a light rod. Find the thrust in the light rod and the reaction at C.

10. Five uniform equal rods AB, BC, CD, DE and EA each of weight W are freely jointed at their ends A, B, C, D and E to form a pentagon. The rods AB and AE make equal angles α and the rods BC and ED make equal angles β with the vertical. The system is hanged from A and the pentagon shape is maintained by connecting B and E by a light rod.

    (i) Find the horizontal and vertical components of the reaction at C.

    (ii) Show that the stress in BE is $W(\tan\alpha + \tan\beta)$.

    (iii) Find the value of the stress when the pentagon is regular.

11. Four equal uniform rods AB, BC, CD and DA each of length $2a$ and weight W are smoothly jointed at A, B, C and D. The midpoints of BC and CD are connected by a light rod of length $2a\sin\theta$. The frame is freely hanged from A.

    (i) Show that the thrust in the light rod is $4W\tan\theta$.

    (ii) Find the reaction at B and C.

12. Four equal uniform rods AB, BC, CD and AD each of weight W are smoothly jointed at their end points to make a square ABCD. The frame is hanged from A. The shape is maintained by joining the midpoints of AB and BC by an inextensible string.

    (i) Show that the reaction at D is horizontal and its magnitude is $\frac{W}{2}$

    (ii) Show that the tension in the string is $4W$

    (iii) Show that the reaction at C is $\frac{W\sqrt{5}}{2}$ and it makes an angle $\tan^{-1}\left(\frac{1}{2}\right)$ with the vertical.

    (iv) Show that the reaction at B is $\frac{W\sqrt{17}}{2}$ and it makes an angle $\tan^{-1}\left(\frac{1}{4}\right)$

13. Four uniform equal rods AB, BC, CD and DA each of weight W are smoothly jointed at the ends to form a square ABCD. The frame is suspended from A and a weight $3W$ is attached to the point C. The shape is maintained by connecting the midpoints of AB and AD with a light rod. Show that the thrust in the light rod is $10W$.

14. Four uniform rods of equal length ℓ and weight W are freely jointed to form a framework ABCD. The joints A and C are connected by a light elastic string of natural length a. The framework is freely suspended from A and takes the shape of a square. Find the modulus of elasticity of the string. Find also the reaction at the joints B and D.

15. Six uniform equal rods each of weight W are smoothly jointed at their end points to form a hexagon ABCDEF. The system is suspended from A and the shape is maintained by light rods BF and CE. Show that the stress in BF is five times the stress in CE.

16. A uniform rod is cut into three parts AB, BC and CD of lengths ℓ, 2ℓ and ℓ respectively. They are smoothly jointed at B and C and rest on a fixed smooth sphere whose radius is 2ℓ and centre O, so that the middle point of BC and the extremities A and D are in contact with the sphere. Show that the reaction on the rod BC at its mid point is $\frac{91W}{100}$ where W is the weight of the rod.

    Find the magnitude and the direction on the rod CD at the joint C and the point whose line of action meets OD.

17. Three uniform rods AB, BC and AC of equal length a and weight W are freely jointed together to form a triangle ABC. The framework rests in a vertical plane on smooth supports at A and C so that AC is horizontal and B is above AC. A mass of weight W is attached to a point D on AB where $AD = \frac{a}{3}$. Find the reaction at joint B.

18. Two uniform equal rods AB and AC each of weight W and length $2a$ are freely jointed at A and placed in a vertical plane with ends B and C on a smooth horizontal table. Equilibrium is maintained by a light inextensible string which connects C to the mid point of AB with each rod making an angle $\alpha\left(<\frac{\pi}{2}\right)$ with the horizontal. Show that the tension T in the string is $T = \frac{W}{4}\sqrt{1+9\cot^2\alpha}$. Find the magnitude and the direction of the reaction at A.

19. Five uniform equal rods each of weight W are smoothly jointed at their ends to form a regular pentagon. CD is placed on a horizontal plane so that the frame is in a vertical plane and the shape is maintained by joining the midpoints of BC and DE by a light rod. Find the reaction at B and show that the tension in the light rod is $\left[\cot\frac{\pi}{5}+3\cot\frac{2\pi}{5}\right]W$.

20. Three equal uniform rods AB, BC, CD each of length $2a$ and weight W are smoothly jointed at B and C, and rest with AB, CD in contact with two smooth pegs at the same level. In the position of equilibrium AB and CD are inclined at an angle α to the vertical BC being horizontal. Prove that the distance between the pegs is $2a\left(1+\frac{2}{3}\sin^3\alpha\right)$. If β is the angle which the reaction at B makes with the vertical, prove that $\tan\alpha \,.\, \tan\beta = 3$.